MULTI-CLASS CLASSIFICATION NEURAL NETWORK REPORT

I. Dataset

The **Car Evaluation Dataset**, sourced from the UCI Machine Learning Repository, was used for Multi-class classification using Neural Network. It has **1,728 instances** of data and **six categorical predictive variables** that include price of **buying, cost of maintenance, number of doors, seat, luggage boot size, and safety.** These attributes are used to classify the acceptability of a car which can be either **unacceptable (unacc), acceptable (acc), good, or very good (vgood)**. The dataset is generated from a decision model and there are no missing data and the data is quite relevant to the actual consumer evaluation criteria. Because of its clearly defined classes and the structured data, it can be employed as a suitable dataset for training neural networks for multi-class classification and finding the tendencies in the processes of vehicles assessment and decision-making.

II. Neural Network Result

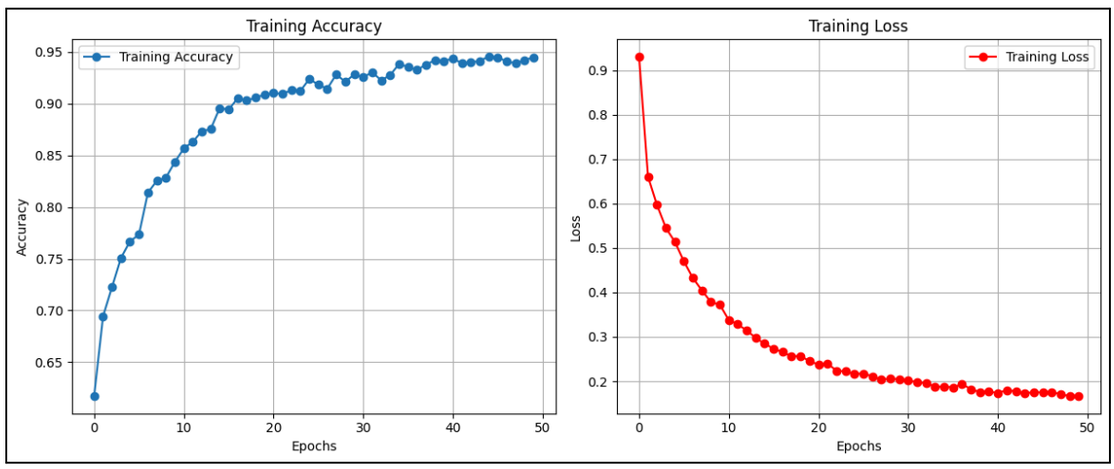


Figure 1: *Training Loss and Accuracy*

A neural network model was implemented for multi-class classification using the TensorFlow and Keras frameworks. The architecture consists of the following layers:

1. **Input Layer:** The model accepts six input features, corresponding to the dataset attributes.
2. **Hidden Layer:** The first dense layer consists of multiple neurons (10) with activation function **relu**, which allows the model to learn complex patterns within categorical features.
3. **Output Layer:** The final layer consists of four output neurons with a softmax activation function, which is appropriate for multi-class classification problems. The four output nodes represent the four different car acceptability categories.

The training process was carried out over 50 epochs and also applying the Regularization early stopping. The **training accuracy curve (blue)** shows a steady

increase, reaching **92.77% accuracy**, demonstrating the model's ability to learn patterns within the dataset effectively. Similarly, the **training loss curve** (in red) steadily decreases, indicating improved model performance with each epoch.

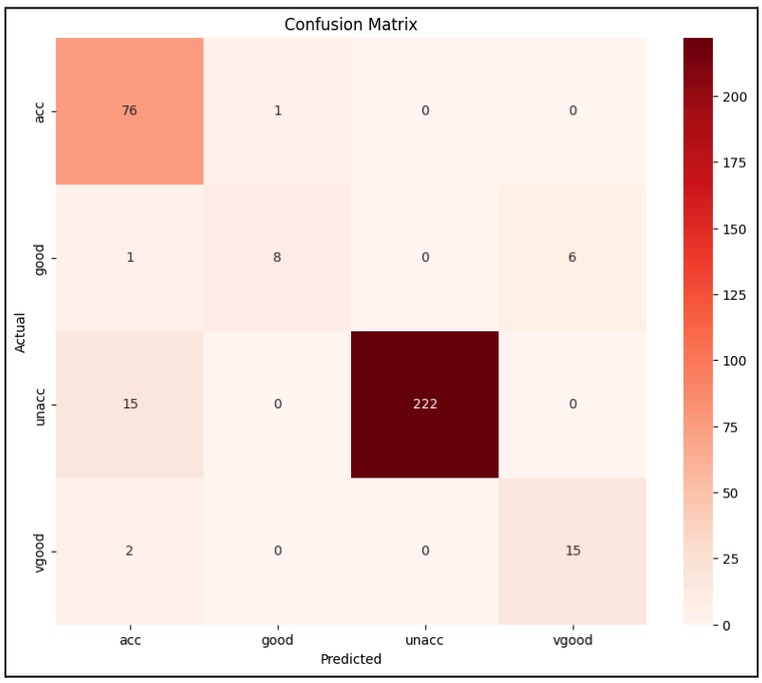


Figure 2: *Confusion matrix*

From the confusion matrix, the majority of the predictions are correctly classified along the diagonal, especially for the unacceptabe (unacc) class which has the highest true positive count of 222. However, there are some misclassifications which are seen especially in the acc class where 15 instances were labeled as unacc and one acc was labeled as good. There is a very low rate of misclassification in the good and very good classes, but in general, the model performs well, especially in marking cars as unacceptable, which might suggest focus on accuracy when working with the most frequent class.

III. References

[1] Marko Bohanec. 1988. Car Evaluation. https://doi.org/10.24432/C5JP48